Opeyemi Jimoh

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EDUCATION

Bachelor of Science in Mechanical Engineering
Purdue University | West Lafayette, Indiana — Expected December 2028
GPA: 3.07
Relevant Coursework: Dynamics, Thermodynamics I, Statics, Intro to Mfg for Mechanical Design, ECE Fundamentals I and Lab

EXPERIENCE

Project Engineering Co-op
Hasbro, Inc. | Pawtucket, Rhode Island — June 2026 – Present
- Designed a universal SLA-printed Nerf blaster accessory in SolidWorks, validated and tested to withstand 100 attach/detach cycles
- Developed the end-to-end manufacturing process for a made-to-order customizable product, spanning injection molding, 3D printing, modular assembly, and distribution
- Executed competitive teardown engineering analysis, characterizing injection-molded components to construct an Excel per-unit cost model
- Delivered cross-functional presentations to Nerf marketing and product design teams to drive material cost reduction

Undergraduate Researcher
CODELab | West Lafayette, Indiana — January 2026 – May 2026
- Designed a dual-net modular vertical recovery system in Siemens NX for a semi-autonomous flapping-wing UAV, generating 3D CAD assemblies and mechanical design artifacts to mitigate structural damage
- Derived dynamic deceleration thresholds and energy absorption limits for incoming flight speeds up to 15 m/s (1.35 kg mass), establishing net compliance criteria to protect 3D-printed airframe components
- Evaluated capture architectures using morphological, Pugh, and weighted decision matrices, selecting a modified COTS frame over a custom pipe structure to cut prototype BOM costs by 35%

Manufacturing Engineer
Purdue Undergraduate Rocket Propulsion Lab | West Lafayette, Indiana — January 2026 – May 2026
- Configured 3-axis CNC toolpaths in Fusion 360 CAM for compressor shroud prototypes, defining tool selections, stock setups, and optimal feed/speed parameters across roughing and finishing operations
- Conducted virtual toolpath simulations and collision checks to verify cutter clearances, optimize tool engagement, and ensure part geometry adhered to 0.005" machining tolerances and GD&T (ASME Y14.5) requirements

Excavation Systems Engineer
Purdue Lunabotics | West Lafayette, Indiana — August 2024 – May 2025
- Evaluated enclosure materials (open mesh vs. laminated woven nylon) for dust protection, executing sandpaper abrasion trials and puncture testing to prevent regolith simulant infiltration into rover drive components
- Diagnosed tensile tear-out failures at riveted mounting interfaces, integrating backup washers and folded edge hems to distribute lateral shear stresses along the rigid frame

PROJECTS

Analog Audio Equalizer
ECE 20007 Final Project — November 2025 – December 2025
- Engineered and prototyped an analog audio equalizer that splits the input signal into bass, mid, and treble bands with independent gain and drives an 8-ohm speaker
- Designed a circuit in LTspice around LM386 op-amps for band filtering, an LF356 for buffering and summation, and an LM386 power amplifier to drive the load, with a master volume stage for output control
- Bench-tested with an oscilloscope and multimeter to verify frequency response, achieving band cutoffs within 10 percent of design tolerance and 585 mW output power

Aquila
Boeing Flight Competition — January 2025 – March 2025
- Developed a balsa-wood glider to support an 11.34g payload, utilizing Excel-based mathematical modeling to optimize the Center of Gravity location relative to the Center of Pressure for a positive static margin
- Modeled aerodynamic performance through 5+ design cycles, achieving a 77% range increase by optimizing wing planform and reducing parasitic drag
- Designed custom assembly fixtures in Fusion 360 and executed rapid FDM 3D printing to enforce tight wing-spar tolerances, eliminating roll-instability identified during flight testing

SKILLS

Software & Programming: Aras Innovator, Claude, ChatGPT, Gemini, SolidWorks, Siemens NX, Fusion 360, Onshape, LTspice, PrusaSlicer, Python, MATLAB, Microsoft Office Suite
Fabrication & Testing: SLA & FDM 3D Printing, 3-Axis CNC Machining & Toolpathing, Manual Mill & Lathe, Sheet Metal Forming, Oscilloscope, Multimeter, Soldering
Engineering Methods: Design for Manufacturability and Assembly (DFM/DFA), GD&T (ASME Y14.5), Injection Molding Teardown/Cost Analysis, Failure Mode Diagnosis, Bill of Materials (BOM)

EXTRACURRICULAR ACTIVITIES

Telecommunications Chair, National Society of Black Engineers (NSBE), Purdue Chapter — May 2025 – May 2026
Member, Purdue Photography Club — August 2025 – Present